class corresponding to the cumulative frequency (cf) just equal or greater than (N/4) is called Q_1 class and the value of Q_1 is obtained by the following formula:

$$Q_1 = L + \left(\frac{\frac{N}{4} - pcf}{f} \right) \times c$$

where L is the lower limit of the Q_1 class,

f is the frequency of the Q_1 class,

c is the magnitude of the Q_1 class,

pcf is the cumulative frequency of the pre Q_1 class.

Similarly third quartile value can be obtained by the same procedure with Q_3 class by the following formula:

$$Q_3 = L + \left(\frac{\frac{3N}{4} - pcf}{f} \right) \times c$$

where L is the lower limit of the third quartile class,

f is the frequency of the third quartile class,

c is the magnitude of the third quartile class,

pcf is the cumulative frequency of the pre Q_3 class.

Similarly the same procedure is to be followed for other positional measures such as deciles and percentiles,

$$D_4 = L + \left(\frac{\frac{4N}{10} - pcf}{f} \right) \times c$$

$$P_{60} = L + \left(\frac{\frac{60N}{100} - pcf}{f} \right) \times c$$

Example 8.4

Compute upper Quartiles, lower Quartiles, D_4 and P_{60}, P_{75} from the following data.

CI	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
Frequency	12	19	5	10	9	6	6

Solution:

CI	f	cf
10 – 20	12	12
20 – 30	19	31
30 – 40	5	36
40 – 50	10	46
50 – 60	9	55
60 – 70	6	61
70 – 80	6	N= 67
	N = 67	

Table : 8.2

Q_1= Size of $\left(\frac{N}{4}\right)^{th}$ value = $\frac{67}{4}$ = 16.75th value.

Thus Q_1 lies in the class (20 – 30) and its corresponding values are $L = 20$;

$\frac{N}{4}$ =16.75; pcf =12; f =19 ; c = 10

$$Q_1 = L + \left(\frac{\frac{N}{4} - pcf}{f} \right) \times c$$

$$Q_1 = 20 + \left(\frac{16.75 - 12}{19} \right) \times 10 = 20 + 2.5 = 22.5$$

Q_3 = Size of $\left(\frac{3N}{4}\right)^{th}$ value = 50.25th value

So Q_3 lies in the class (50-60) corresponding values are L = 50, $\left(\frac{3N}{4}\right)$ = 50.25; pcf = 46, f = 9, c = 10

$$Q_3 = L + \left(\frac{\frac{3N}{4} - pcf}{f} \right) \times c$$

$$Q_3 = 50 + \left(\frac{50.25 - 46}{9} \right) \times 10 = 54.72$$

$$D_4 = L + \left(\frac{\frac{4N}{10} - pcf}{f} \right) \times c$$

D_4 = Size of $\left(\frac{4N}{10}\right)^{th}$ value = 26.8th value. Thus D_4 lies in the class (20 – 30) and its corresponding values are

$L = 20$, $\frac{4N}{10}$ = 26.8; pcf = 12, f =19, c = 10.

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